

MVE441 Exam Report (Data set X8)

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June 2026

1 Question 1(a)

An exploratory analysis was conducted to inform the classification pipeline. The data set is high-dimensional (820 features, 8080 observations), with little feature redundancy according to the correlation analysis. PCA on standardized features showed that variance is distributed across many components (figure 1a), although the 2D projection still revealed visible class clustering (figure 1b), suggesting meaningful class structure.

The labels were moderately imbalanced, with class sizes ranging from 489 (class 4) to 1630 (class 7), motivating the use of balanced accuracy and macro-F1 rather than overall accuracy. Feature scales also varied substantially, supporting feature standardization for scale-dependent methods.

Based on these observations, three classifiers were evaluated: logistic regression (with L2 penalty), a linear SVM, and a random forest. The linear models provided strong baselines for high-dimensional data, while the random forest was included to capture potential non-linear relationships. Scale-dependent models were fit using standardized features, and all classifiers used class-balanced weights.

Performance was assessed using repeated stratified 5-fold cross-validation with three random seeds. Hyperparameters were selected through an inner stratified cross-validation loop, and models were compared using balanced accuracy, macro-F1, and class-specific recall.

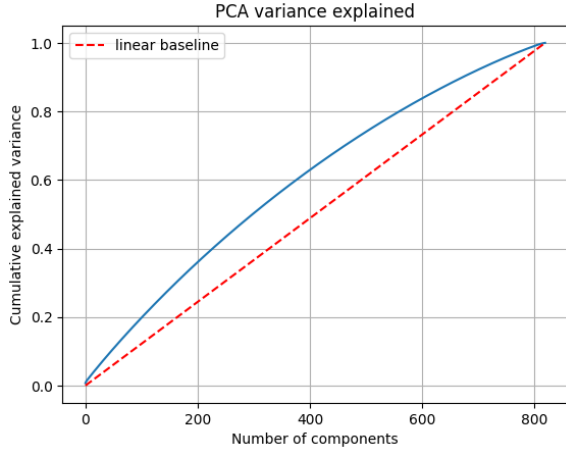
To investigate the effect of sample size, the analysis was repeated using stratified subsamples of 1000, 3000, and 8080 observations. Performance improved for all models as sample size increased (figure 2), although the magnitude of improvement differed. Random forest achieved the highest balanced accuracy and macro-F1 at all sample sizes, reaching approximately 0.94 on the full data set, suggesting that important non-linear structure exists in the data. Logistic regression consistently outperformed the linear SVM and showed the largest gains with increasing sample size. The similarity between balanced accuracy and macro-F1 indicates that precision and recall were reasonably balanced across classes despite the moderate class imbalance. Performance variability also decreased with larger sample sizes, indicating more stable estimates.

Class-specific recall revealed differences in classification difficulty. As seen in figure 3, class 6 was the most challenging for the linear classifiers. Random forest achieved the highest recall across all classes and showed the largest improvements on difficult classes, further supporting the presence of non-linear class boundaries. While logistic regression outperformed SVM, the gap between both linear methods and random forest suggests that linear decision boundaries alone are insufficient to fully capture the class structure.

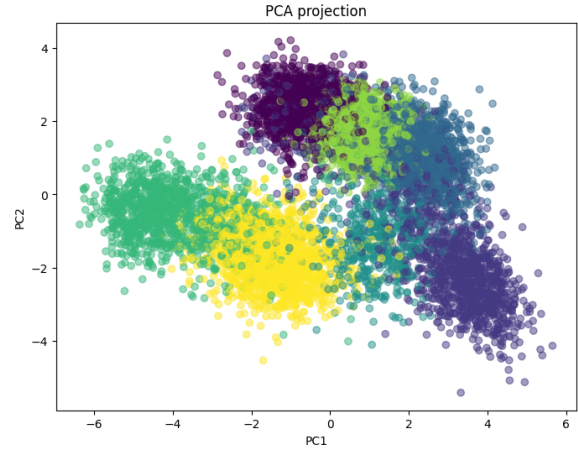
Feature informativeness was assessed using method specific importance measures. Predictive information appeared to be distributed across many variables as logistic regression (with L1 penalty) retained approximately 633 of 820 features, while random forest required about 437 features to explain 80% of cumulative importance. Consistent with the PCA results, this suggests that the data set lacks a small set of dominant predictors and that the number of informative features depends on the model and importance criterion used.

1.1 Conclusions

- The data set contains structure which allows for high accuracy class predictions to be made for the classifiers, but not in a sparse form. PCA and feature importance analysis both suggest that predictive information is spread out across many features.
- Random forest performed best overall across the classes with the highest balanced accuracy, macro F1 score and class-specific recall, indicating that class structure is to some degree nonlinear and as such benefits from a more flexible classifier.
- Logistic regression consistently outperformed linear SVM, showing that the data does contain useful linear information. However, the gap between logistic regression and random forest shows that linear decision boundaries cannot fully capture the class separation.
- Specific class prediction difficulty was uneven, as class 6 was the hardest class for the linear methods, while for example class 5 and 7 were much easier. These differences were more leveled-out for random forest.

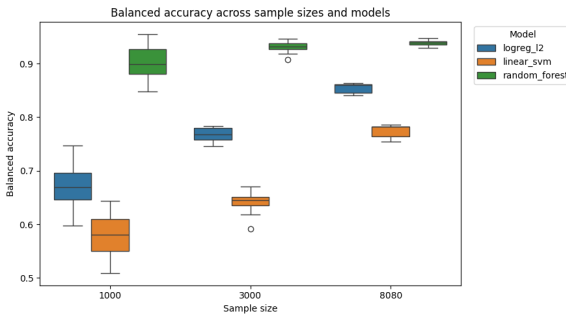


(a) Cumulative explained variance compared to number of components.

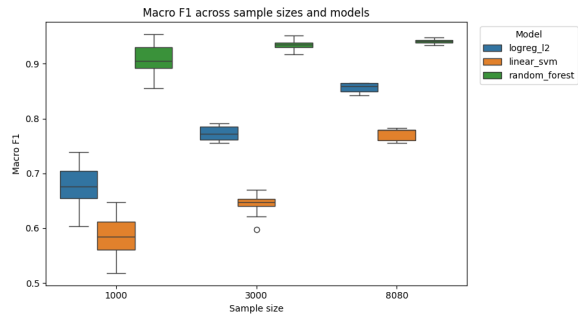


(b) Data projected onto the first two principal components.

Figure 1: Figure (a) shows a near linear (red) relationship between the cumulative explained variance and number of components (blue) but (b) shows that clustering does occur for the different classes.



(a) Boxplot of balanced accuracy.



(b) Boxplot of macro F1 score.

Figure 2: The balanced accuracy and macro F1 score in figure (a) and (b) respectively for the classifiers used in the pipeline, evaluated at sample sizes 1000, 3000 and 8080. The results show that both metrics are monotonically increasing for increased sample sizes and share overall similar values, indicating that precision and recall are relatively balanced across classes.

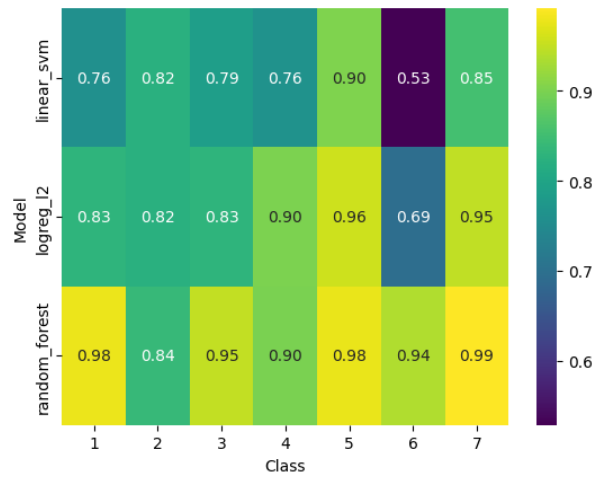


Figure 3: Mean class-specific recall for the classifiers evaluated on the full train data (8080 observations). The results show class 6 being significantly harder to classify correctly for the linear classifiers, as compared to random forest, implying linear boundaries may not capture the full class structure. The class imbalance does not appear to significantly impair results, a consequence of using balanced weights and stratified data sets.

2 Question 1(b)

To create a minimum disruption tailored to this data set the three most important features according to random forest importance was removed from the data. This choice was motivated by the earlier analysis, which showed that the predictive signal is spread across many variables rather than concentrated in only a few. After some exploratory attempts with stronger disruptions, removing only the top three features was selected as the smallest disruption that still produced a measurable and repeatable performance drop. The same repeated stratified cross-validation pipeline as in 1 (a) was then applied to both the clean and disrupted data, so the comparison is based on the same evaluation protocol across multiple seeds.

Results in table 1 show that the disruption affected all three classifiers. Balanced accuracy and macro F1 decreased for every method. This suggests that the removed features contained predictive information that was useful for both linear and nonlinear decision boundaries. Rather than targeting a mechanism unique to random forest, the disruption appears to have removed information that was broadly important throughout the dataset, which explains why all methods experienced a similar degradation in performance. Random forest remained the best performing classifier after the disruption, but its performance still dropped from 0.94 to 0.875 in balanced accuracy. Logistic regression and SVM were also weakened with balance accuracy dropping from 0.844 to 0.775 and from 0.758 to 0.711 respectively. The class wise recall plots in figure 4 show that the reduction was not uniform across classes, the harder classes lost more recall than the easier ones, which is consistent with the removed features contributing to several class boundaries rather than to one isolated decision rule.

The fact that all methods deteriorated rather than only random forest suggests that the removed features were not specific to random forest alone. Instead, they appear to carry broad predictive information that is useful for both linear and nonlinear classifiers. This is also consistent with the earlier finding that informative signal is distributed across many features. As a result, even a small disruption of removing the three top-ranked features was enough to weaken all models, but not enough to cause a collapse. That makes the disruption minimal in the sense required by the question, even if it does not isolate a single method as strongly as hoped.

2.1 Conclusions

- Removing only the three most important random forest features was sufficient to produce a consistent reduction in performance across all classifiers, demonstrating that even a small targeted disruption can affect the dataset.
- All models experienced decreases in balanced accuracy and macro F1, indicating that the removed features contained predictive information that was broadly useful rather than specific to a single classifier.
- Random forest remained the strongest performing method after disruption, although it also suffered a substantial reduction in performance, showing that no method was completely resilient to the feature removal.
- The class-wise recall analysis showed that the disruption affected some classes more than others, suggesting that the removed features contributed to multiple class boundaries rather than a single classification task.
- Overall, the experiment demonstrates that the dataset is sensitive to the removal of a few highly informative features, while also showing that the predictive signal is sufficiently distributed that no single small disruption causes a complete failure of any method.

Model	Dataset	Balanced Accuracy	Macro F1	Accuracy
Linear SVM	Clean	0.758	0.754	0.763
Linear SVM	Disrupted	0.711	0.706	0.715
Logistic regression	Clean	0.844	0.846	0.844
Logistic regression	Disrupted	0.775	0.775	0.780
Random forest	Clean	0.940	0.942	0.943
Random forest	Disrupted	0.875	0.885	0.904

Table 1: Mean performance on the clean and disrupted data for the three classifiers. The disrupted data were obtained by removing the three most important random forest features. Values are averaged across repeated stratified cross-validation runs.

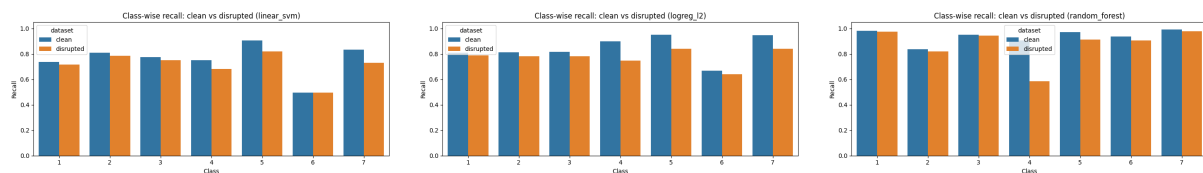


Figure 4: Class wise recall on the original data and disrupted data after removing the three most important random forest features on SVM (left), logistic regression (middle) and random forest (right). Recall decreased for classes across all three classifiers, although the effect varied between classes. This indicates that the removed features contained predictive information that was broadly useful to both linear and nonlinear models, resulting in a measurable but limited performance decline.

3 Question 1(c)

In order to look for evidence of mislabeling in the data set I used cross-validation with random forest to check out of fold predictions in order to give each observation a suspicion score. This suspicion score was based on both the average predicted probability for the observed label and repeated disagreement across folds. The logic is that observations whose labels are incorrect will tend to receive low predicted probability for their assigned class and be repeatedly classified as another class across different resamples. Several observations were consistently flagged as highly suspicious, with the top-ranked observations showing very low average predicted probability for the correct class and disagreement in every resample as seen in table 2. The observations with highest suspicion scores were found mostly in class 2, and this pattern was also found by repeating this process using logistic regression, which further supports that these are either unusual or possibly mislabeled observations rather than model specific shortcomings.

To verify that the detection method works, I randomly mislabeled 5% (404) of the labels and reran the same random forest detecting method. Out of the 404 mislabeled observations, 321 of them were detected in the 404 most suspicious observations, as can be seen in table 3. Figure 5 provides further evidence for the effectiveness of the approach where we see that, while the overall shape of the suspicion score distribution remains similar after label injection, the mislabeled data set exhibits a noticeably heavier right tail, with many more observations receiving suspicion scores close to 1.0. These observations correspond to cases where the model consistently disagrees with the assigned label and assigns it very low probability, which is precisely the behavior expected for mislabeled samples. Together, the recovery rate and the shift in the suspicion score distribution support the validity of the detection method.

3.1 Conclusions

- The method used identified a small subset of suspicious observations, with consistent low class probabilities and disagreement amongst predicted label across different runs. Thus indicating inconsistency compared to the learned decision boundaries on the overall data.
- Suspicious observations were not uniformly distributed amongst the classes, but rather they were mostly found in class 2. This suggests that either there is in-class ambiguity in the class which makes it harder to classify or that potential mislabeling.
- Validation through injected mislabeling in the data showed that the detection technique successfully detected 321 out of the 404 intentionally mislabeled observations, showing high accuracy for the detection technique.

Index	True label	Mean Self Pred. Prob.	Disagreement	Suspicion Score
5349	2	0.060	1.0	0.970
606	2	0.064	1.0	0.968
7696	2	0.067	1.0	0.967
797	2	0.067	1.0	0.967
344	2	0.069	1.0	0.966

Table 2: Top-ranked suspicious observations in the original data set based on cross-validation out of fold random forest predictions. These points consistently receive near zero probability for their observed label and are misclassified across all folds.

	Injected	Recovered (top 404)	Recovery rate
Random 5% label noise	404	321	0.795

Table 3: Validation of the mislabel detection method. A random set of 5% of labels were corrupted and the detector was rerun. The recovery rate is far above chance, indicating that the method successfully identifies injected mislabels.

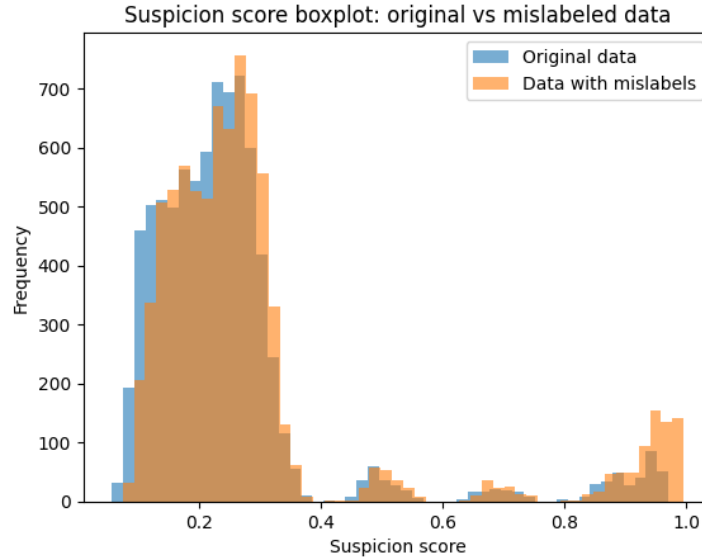


Figure 5: Distribution of suspicion scores before and after injecting 5% random label noise. The injected mislabels produce a heavier right tail, indicating that mislabeled observations are more likely to receive high suspicion scores.

4 Question 1(d)

The two best performing methods from 1 (a), logistic regression and random forest, were trained on stratified subsamples of sizes 1000, 3000 and 8080 observations as previously in 1 (a). In order to provide confidence to the predictions on X_{TE} I looked at the mean maximum predicted class probability and the probability margin, meaning the difference between the largest and second-largest class probabilities. These metrics were first evaluated on out of fold predictions in the cross-validation in X_{TR} where higher probability values indicated more correctness, as can be seen in figure 6 where there is a clear distinction in the mean maximum class probabilities when predictions are correct or wrong for the two classifiers. The fact that logistic regression produced higher predicted probabilities than random forest is expected as logistic regression optimizes log-loss and is typically well calibrated, compared to random forest where probabilities are obtained by averaging class fractions from individual tree leaves and are often more conservative and has less sharp peaks.

Similar balanced accuracy and macro F1 results were found as in 1 (a) (no plot is provided here as they are practically identical to that in 1 (a)) where random forest performs the best and performance grew with sample size. Confidence also increased with sample size (figure 6) where logistic regression became very certain for both correct and incorrect predictions. This indicates that additional training data improve both predictive performance and the stability of the probability outputs.

Applied to X_{TE} , the mean maximum probability and probability margin also increased with training sample size (figure 7), suggesting that predictions become more certain as more training data are available. For the full training set, logistic regression achieved a mean maximum predicted probability of approximately 0.93 and a mean probability margin of 0.88, while the corresponding values for random forest were approximately 0.55 and 0.38. Although random forest produced lower probability estimates, its superior cross-validation performance indicates that this reflects differences in probability calibration rather than lower predictive quality. Since higher confidence was associated with higher correctness in the out of fold analysis, these results provide evidence that the predictions on X_{TE} are reasonably reliable, particularly for models trained on the full data set.

Using the X_{TE} data, the two models produced stable and mostly similar class distributions, as seen in figure 8. Class 4 was consistently the least frequently predicted class. The similarity of the class distributions across methods for the different sample sizes suggests that the dominant class structure identified in X_{TR} is also present in X_{TE} . Some differences were observed, with random forest assigning more observations to class 3 and fewer to class 2 than logistic regression, indicating that the models capture somewhat different decision boundaries. However, these differences were modest relative to the overall agreement between the methods.

In order to assess the existence of drift, meaning whether X_{TE} differs from the training distribution, a domain classification experiment was performed. The observations from X_{TR} and X_{TE} were combined and assigned labels indicating whether they originated from the training or test set. Both logistic regression and random forest were then trained using cross-validation to distinguish between the two data sets. The logic is if there exists substantial drift then the classifiers should be able to separate the data sets and achieve balanced accuracy above chance level.

In table 4 we see that logistic regression got a balanced accuracy of 0.519 and random forest got 0.5. Since both these values are essentially at chance levels this indicates that the classifiers were basically unable to reliably distinguish the data sets. As such we do not observe any strong indication of drift in X_{TE} .

4.1 Conclusions

- Logistic regression and random forest both produced usable confidence estimates on X_{TE} and the out of fold results on X_{TR} showed that higher predicted probability were associated with higher correctness.
- Confidence increased with sample size, indicating that more training data improved certainty around predictions.
- The predicted class distributions for X_{TE} were similar across the two models for different sample sizes, indicating that the distribution of X_{TE} is close to what was found in X_{TR} .
- The domain classification test gave balanced accuracy values at chance levels and as such there is little evidence that X_{TE} differs substantially from X_{TR} .
- Overall, the final predictions on X_{TE} appear reasonable, especially when the models are trained on the full data set, and there is no sign of drift that would make the results less reliable.

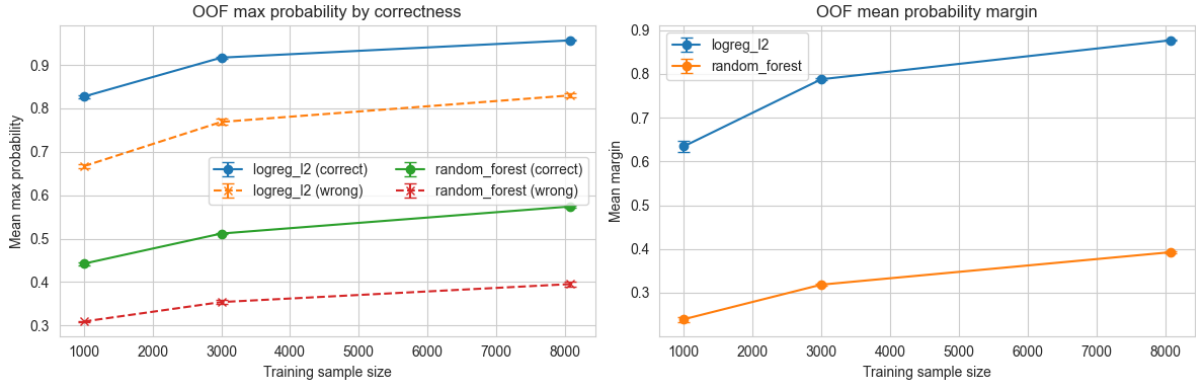


Figure 6: Out of fold prediction confidence as a function of training sample size for logistic regression and random forest. The left figure shows the mean maximum predicted class probability for correct and incorrect predictions, while the right figure shows the mean probability margin. Correct predictions consistently receive higher confidence than incorrect predictions for both methods, supporting the use of these metrics as indicators of prediction reliability. Confidence increases with sample size, with logistic regression producing substantially higher probability estimates than random forest.

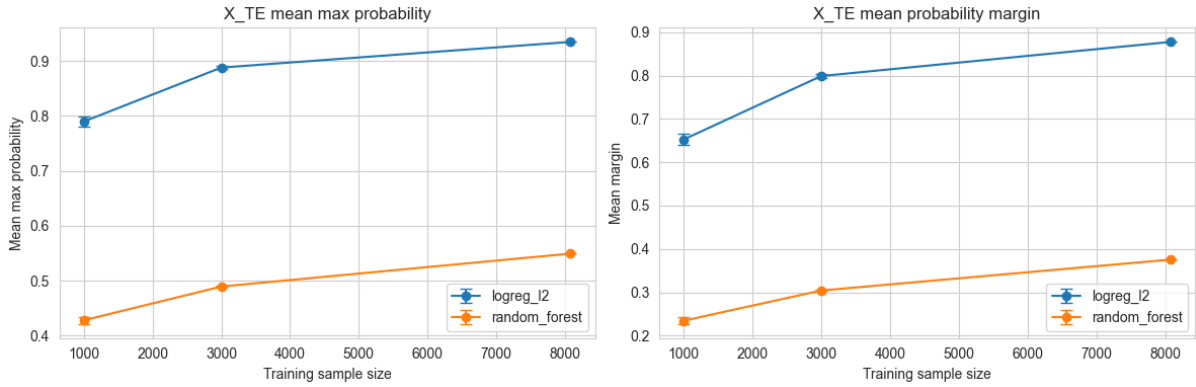


Figure 7: Mean maximum predicted probability (left) and probability margin (right) on X_{TE} for different training sample sizes. Confidence increases with sample size for both methods, while logistic regression produces substantially higher probability estimates than random forest. The probabilities are similar to those produced on X_{TR} .

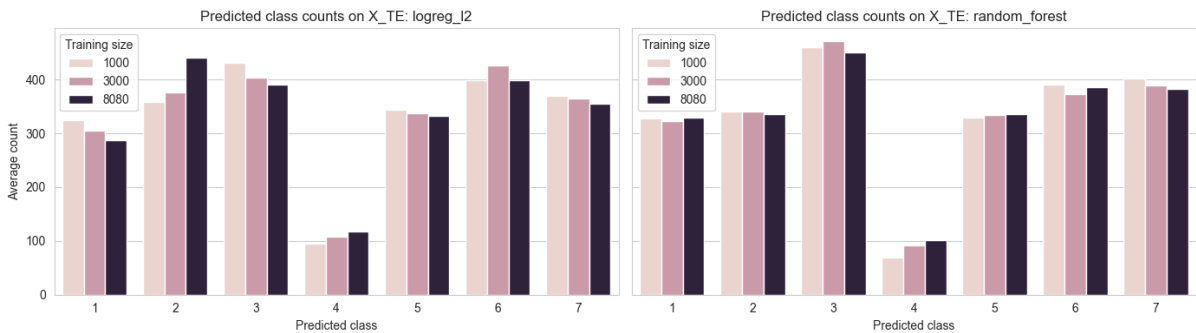


Figure 8: Predicted class counts on X_{TE} across training sample sizes for logistic regression and random forest. The class distributions are highly consistent across methods and sample sizes, suggesting similar underlying class structure in X_{TE} .

Classifier	Balanced Accuracy	Standard Deviation
Logistic Regression	0.519	0.006
Random Forest	0.500	0.000

Table 4: Balanced accuracy from domain classification used to distinguish observations from X_{TR} and X_{TE} . Performance is close to chance level for both methods, providing little evidence of substantial drift in X_{TE} .

5 Question 2(a)

To create a more challenging problem, the data was rebalanced such that class 7 became a super-majority class with 1000 observations and the remaining classes kept only 50 observations each. This extreme imbalance makes choice of classifiers crucial as overall accuracy will give optimistic results because of the super-majority class. As such balanced accuracy and macro F1 were the main metrics for the three classifiers used in 1 (a) which were trained using three different strategies: plain training on the data, class weighted training and oversampling in which for the small classes duplicates of observations were made and added to the data until each class had 1000 observations.

The results shown in table 5 and figure 9 show that the imbalance severely affects performance on the minority classes. For logistic regression, both class weighting and oversampling improved performance relative to the plain model and less drastic but similar results were found for SVM as well. As shown by the class wise recall patterns in figure 10, the imbalance strategies generally improved class recall for the minority classes while preserving performance on the majority class, suggesting that the linear models benefited from the acknowledgment and acting on the imbalance, since minority classes receive more influence during training.

Random forest behaved very differently compared to the linear models. The random forest model with plain data performed the best overall (table 5 and figure 9) compared to the other plain models, reaching balanced accuracy of about 0.6 and macro F1 0.69, while class weighting and oversampling collapsed the performance. In the class weighted case random forest effectively turned into only guessing the majority class. This is not what I expected but could perhaps indicate that the weighting and oversampling schemes interacted unfavorably with the tree-building process under this extreme case.

Overall the experiment shows that extreme imbalance can strongly distort model behavior and that the best correction depends on the classifier and the severity of the imbalance.

5.1 Conclusions

- Extreme imbalance made the problem much harder for the minority classes, even though overall accuracy stayed high because class 7 dominated the data. As such balanced accuracy and macro F1 were much more informative as they show the performance gap for the minority classes compared to the supermajority class.
- Logistic regression benefited most from imbalance correction and both class weighting and oversampling improved balanced accuracy, macro F1, and class recall for the minority classes.
- Linear SVM improved only slightly, suggesting that it was less affected by the imbalance correction than logistic regression.
- Random forest performed best without any correction and class weighting and oversampling made it much worse, with minority class recall collapsing in the weighted case. This shows, if the results are representative of the actual model behavior, that imbalance techniques do not help all classifiers equally.

Model	Strategy	Balanced accuracy	Macro-F1	Accuracy
Linear SVM	Plain	0.581	0.584	0.845
Linear SVM	Weighted	0.583	0.586	0.846
Linear SVM	Oversample	0.586	0.600	0.858
Logistic regression	Plain	0.528	0.583	0.871
Logistic regression	Weighted	0.643	0.681	0.899
Logistic regression	Oversample	0.631	0.672	0.896
Random forest	Plain	0.603	0.690	0.893
Random forest	Weighted	0.148	0.134	0.771
Random forest	Oversample	0.233	0.277	0.793

Table 5: Mean classification performance under the extreme imbalance setting with class 7 as the supermajority and 50 observations in each minority class. Balanced accuracy and macro-F1 reveal large differences between strategies that are not visible in ordinary accuracy.

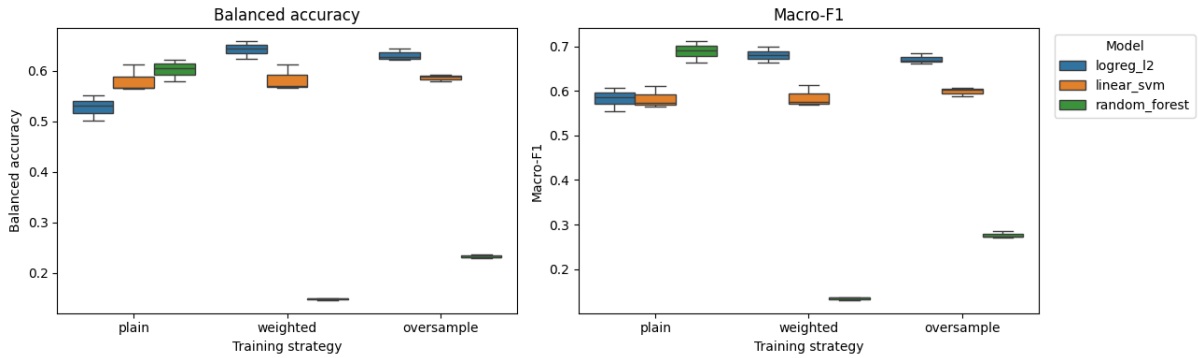


Figure 9: Balanced accuracy (left) and macro F1 (right) obtained under the extreme class-imbalance scenario where class 7 contains 1000 observations and all remaining classes contain 50 observations. Results are shown for logistic regression, linear SVM and random forest trained using plain training, class weighting and oversampling. Logistic regression and SVM benefit from imbalance-aware training, with oversampling generally producing the strongest results. In contrast, random forest performs best without imbalance correction.

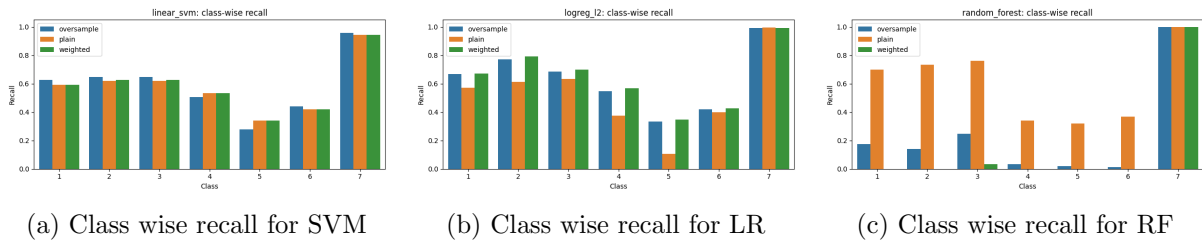


Figure 10: Class wise recall for (a) linear SVM, (b) logistic regression and (c) random forest under the extreme imbalance setting. Each color corresponds to a different imbalance handling strategy (blue - oversample, orange - plain, green - weighted). For the linear models, weighting and oversampling improve recall for minority classes while maintaining high recall for the majority class. Random forest shows the opposite behavior.

6 Question 2(b)

The other approach tried to handle the imbalance by using a hierarchical classification strategy. In the first stage, logistic regression was trained to separate class 7 from all remaining minority classes. In the second stage random forest was used trained on the remaining classes 1 to 6. Logistic regression was chosen for the first stage because it handled imbalance relatively well in 2 (a) and is well suited for the binary majority-vs-rest problem. Random forest was chosen for the second stage as it was the best performing classifier in 1 (a) and can model the more complex decision boundaries among the minority classes.

The hierarchical pipeline performed well at both stages. As shown in table 6, logistic regression achieved a balanced accuracy of 0.848 and a macro F1 of 0.932 when separating class 7 from the other classes. The second stage random forest achieved a balanced accuracy of 0.881 and macro F1 0.879 on the minority classes. This indicates that the supermajority class can be identified reliably and that the remaining classes become much easier to separate once class 7 was removed.

Figure 11 compares the overall hierarchical classifier with the plain random forest results from 2 (a), where we see that the hierarchical approach improved balanced accuracy from 0.603 to 0.687 and slightly improved macro F1. However, overall accuracy decreased from 0.893 to 0.868, which was to be expected as the plain random forest classifier benefits from the dominance of class 7 and therefore gains accuracy by more heavily predicting the majority class.

The class wise recall results in figure 12 show the main advantage using the hierarchical model as several of the minority classes saw an increase in recall, all the while the recall for class 7 remained high. Although recall decreased for some minority classes the hierarchical classifier provided a substantially more balanced treatment of the classes.

6.1 Conclusions

- A hierarchical strategy was effective for this extreme imbalance problem and gave showed better balanced accuracy performance than the plain classifiers.
- Logistic regression was a good choice for the first stage because it handled the majority-vs-rest split reliably and gave strong performance on the binary problem.
- Random forest worked well for the second stage of the hierarchical model because once the supermajority class was removed the multiclass problem became easier and more balanced.
- Compared with the plain random forest from 2(a), the hierarchical model improved balanced accuracy and macro F1, and as such it treated the minority classes more fairly.

Stage	Balanced Accuracy	Macro F1
Stage 1 (Class 7 vs Rest)	0.848	0.932
Stage 2 (Classes 1–6 only)	0.881	0.879
Overall	0.687	0.6857

Table 6: Performance at different stages of the hierarchical classification pipeline. Stage 1 corresponds to logistic regression separating class 7 from the remaining classes, while Stage 2 corresponds to random forest classifying among classes 1–6 only.

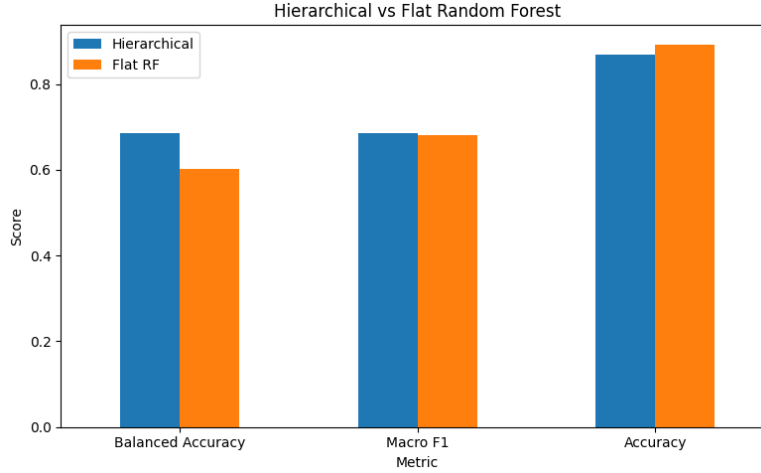


Figure 11: Comparison of the hierarchical classifier and the plain random forest on the extreme class imbalance data set. The hierarchical approach increased balanced accuracy from 0.603 to 0.687 and achieved a small improvement in macro F1, indicating better overall treatment of minority classes. However, overall accuracy decreased from 0.893 to 0.868 because the flat random forest benefited from the dominance of class 7 and therefore achieved higher accuracy by more frequently predicting the majority class.

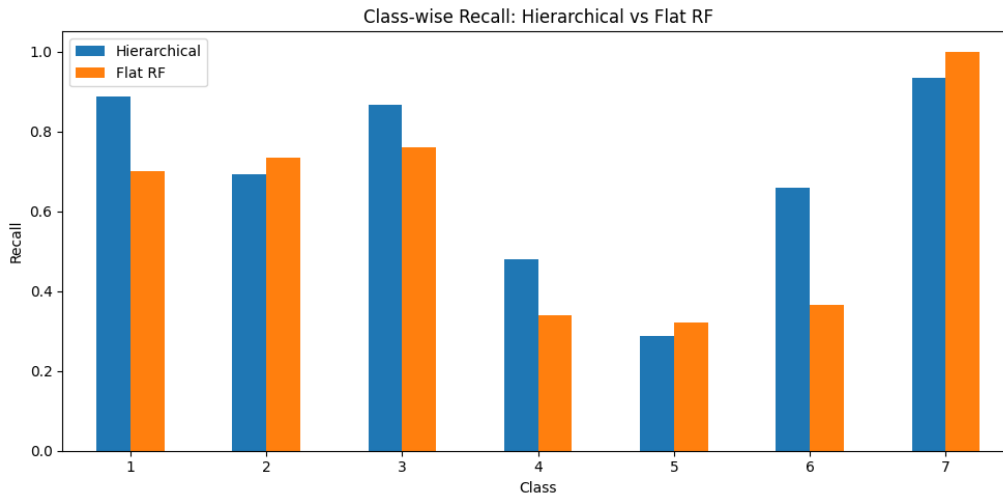


Figure 12: The hierarchical approach improved recall for several minority classes while maintaining high recall for the supermajority class. These results show that separating class 7 in the first stage produced a more balanced classifier, reducing model bias to favor the majority class at the expense of minority class detection.